LEONG MING REN

Lab 2_ Exploring OWASP & Web Security Risk (1)

LAB 02 Sem1 20262027

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UNIVERSITI TUN HUSSEIN ONN MALAYSIA
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LAB 02: EXPLORING OWASP & WEB SECURITY RISK

Role	Student Name	Matric ID	Section
Leader & Member 1	Leong Ming Ren	DI250140	1
Member 2	Teo Jia Mun	DI250139	1
Member 3	Saw You Huang	DI250050	1
Member 4	Muhammad Danish bin Badrul Hisham	CI260111	1

Task 1: OWASP Top 10 Risk Analysis & Case Study

Instructions:

1. **Visit the OWASP official website: https://owasp.org/.**
2. **Provide a brief introduction to OWASP by answering:**
    - **What is OWASP, and why is it important?**

3

OWASP is an open community dedicated to allow organizations to develop, purchase and maintain applications and APIs that can be trusted. It is important as it helps organizations to understand, identify and also prevent security risks in web applications and also software systems.

    - **What is the OWASP Top 10, and how does it help organizations?**

2

The OWASP Top 10 is a security awareness document that is regularly updated by the OWASP, which includes broken access control, security misconfiguration, software supply chain failures, cryptographic failures, injection, insecure design, authentication failures, software or data integrity failures, security logging & alerting failures and mishandling of exceptional conditions. It helps organizations to find the common weaknesses in their web applications, and also provides guidance for developers to develop, test and maintain. Therefore, it helps them to reduce risks such as unauthorized access, data leaks and information theft.

    - **How often is the OWASP Top 10 updated, and why does it change?**

- It is updated regularly, rather than on a fixed annual schedule. The reason for changing OWASP Top 10 regularly is to prevent new techniques that exploit web applications. New attack methods are constantly emerging, some vulnerabilities become more common, OWASP combines feedback from cybersecurity experts with actual vulnerabilities data to identify the most dangerous threats.

3. **Select one security risk from the OWASP Top 10.**

   Security Misconfiguration.

4. **Provide a detailed analysis of the selected risk, including:**
   - **Risk Description: What does this vulnerability mean?**

     It is missing appropriate security hardening across any part of the application stack or improperly configured permissions on cloud services.

   - **How It Happens: How do attackers exploit this weakness?**
     - ➔ The application server's configuration allows detailed error messages, such as stack traces to be returned to users.This potentially exposes sensitive information such as component versions that are known to be vulnerable.
     - ➔ A cloud service provider(CSP) defaults to having sharing permissions open to the Internet. This allows sensitive data stored within cloud storage to be accessed.
     - ➔ The application server comes with sample applications not removed from the production server.
   - **Prevention: How can developers protect web applications from this risk?**
     - ➔ Sending security directives to clients(For example Security Headers).
     - ➔ An automated process to verify the effectiveness of the configurations and settings in all environments..
     - ➔ Do a task to review and update the configurations appropriate to all security notes, updates, and patches as part of the patch management process and Review cloud storage permissions.
     - ➔ Proactively add a central configuration to intercept excessive error messages as a backup.
     - ➔ Segmented application architecture provides effective and secure separation between components with segmentation.

5. **Research a real-world web security breach that reflects your selected risk.**

   Title: Accenture Cloud Storage Data Exposure Incident (2017)
   - ➔ Organization: Accenture, a global consulting and technology services company.
   - ➔ Year: 2017.
   - ➔ Incident: Four Amazon Web Services (AWS) S3 storage buckets belonging to Accenture were discovered to be publicly accessible.
   - ➔ Discovery: Security researcher Chris Vickery from UpGuard discovered the exposed storage buckets in September 2017.

- ➔ Data Exposed: The exposed information reportedly included passwords, cloud access credentials, encryption keys, certificates and internal system information.
- ➔ Security Issue: The incident occurred because the cloud storage permissions were incorrectly configured potentially allowing unauthorized parties to access sensitive information.
- ➔ Resolution: Accenture was notified and the exposed storage buckets were secured.

6. Answer the following in relation to the chosen risk:

- **What happened in the breach?**

  In 2017, Accenture was found to have four AWS S3 storage buckets that were publicly accessible due to incorrect security configurations. These storage buckets contained sensitive information, including passwords, cloud credentials, encryption keys and internal technical data. The exposure was discovered by a security researcher, who notified Accenture so that the affected storage buckets could be secured. This incident highlighted the risk of exposing confidential information through improperly configured cloud storage.

  How does the breach relate to the selected OWASP risk?

  This incident is directly related to Security Misconfiguration, which is listed as A02:2025 in the OWASP Top 10. The AWS S3 storage buckets were configured with permissions that allowed public access to sensitive files. This demonstrates how incorrect security settings can expose confidential information to unauthorized parties. According to OWASP, improperly configured cloud service permissions are an example of Security Misconfiguration.

- **What could have prevented the attack?**

  This incident could have been prevented by configuring AWS S3 storage permissions securely and restricting access to authorized users only. Accenture could also have enabled AWS S3 Block Public Access applied the principle of least privilege and conducted regular security configuration reviews. Automated security checks and continuous monitoring could have detected the publicly accessible storage buckets earlier and reduced the risk of sensitive data exposure. |

7. Reference

1. OWASP. (2023). *OWASP foundation, the open source foundation for application security*. Owasp.Org. https://owasp.org/
2. Kovacs, E. (2017, October 11). *Accenture exposed data via unprotected cloud storage bucket*. SecurityWeek. https://www.securityweek.com/accenture-exposed-data-unprotected-cloud-storage-bucket/